

Swiss Institute of Banking and Finance



University of St.Gallen

6. Corporate Financing & the Issuance of Securities

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3,130/4,130 Corporate Finance

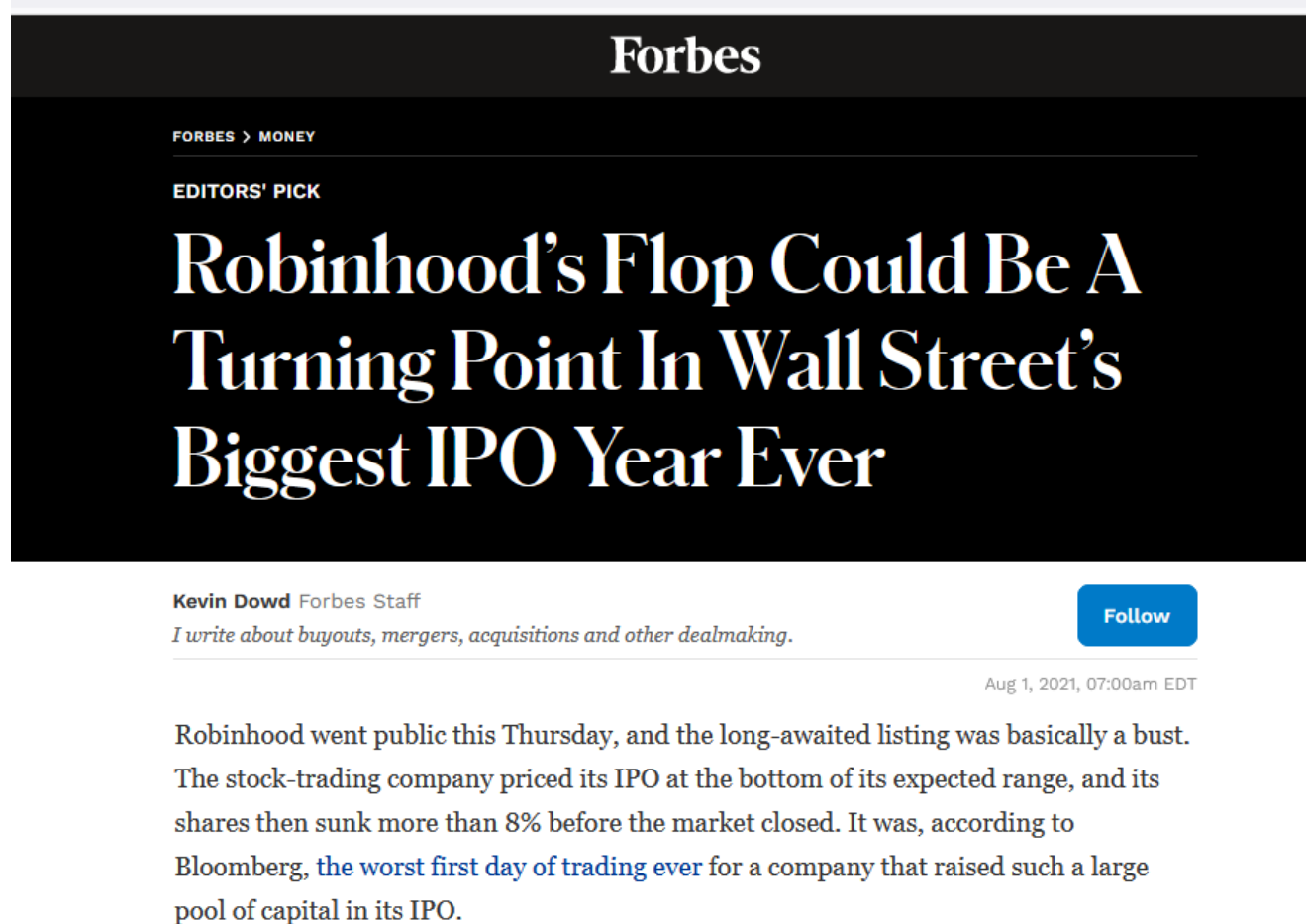
Course Outline

1. General Information
2. Risk and Return
3. Portfolio Theory and CAPM
4. Corporate Valuation
5. Cost of Capital
- 6. Corporate Financing and the Issuance of Securities**
7. Capital Structure
8. Risk Management
9. Conflicts of Interest and Corporate Governance
10. Market Efficiency and Behavioral (Corporate) Finance
11. Empirical Research in Corporate Finance

After this lecture you will have understood...

- ...what **patterns of corporate financing** the companies rely on: Internal vs. external (new issues of debt or equity) financing.
- ...the difference between lenders' and stockholders' **rights**.
- ...the role of **venture capital** organizations and how companies raise venture capital.
- ...the **performance** of venture capital.
- ...the **IPO process**: What is the role of the underwriters? Why are new issues generally underpriced? How do they perform in the long-term?

...which are topics regularly covered by the public press.



The screenshot shows a Forbes article. At the top is the 'Forbes' logo. Below it, the navigation bar shows 'FORBES > MONEY'. The article is marked as an 'EDITORS' PICK'. The title is 'Robinhood's Flop Could Be A Turning Point In Wall Street's Biggest IPO Year Ever'. The author is 'Kevin Dowd Forbes Staff' with a bio: 'I write about buyouts, mergers, acquisitions and other dealmaking.' and a 'Follow' button. The date is 'Aug 1, 2021, 07:00am EDT'. The text of the article begins: 'Robinhood went public this Thursday, and the long-awaited listing was basically a bust. The stock-trading company priced its IPO at the bottom of its expected range, and its shares then sunk more than 8% before the market closed. It was, according to Bloomberg, the worst first day of trading ever for a company that raised such a large pool of capital in its IPO.'

- Other recent IPOs (e.g., Dropbox and Uber) performed poorly after going public and then announced changes to their business model.

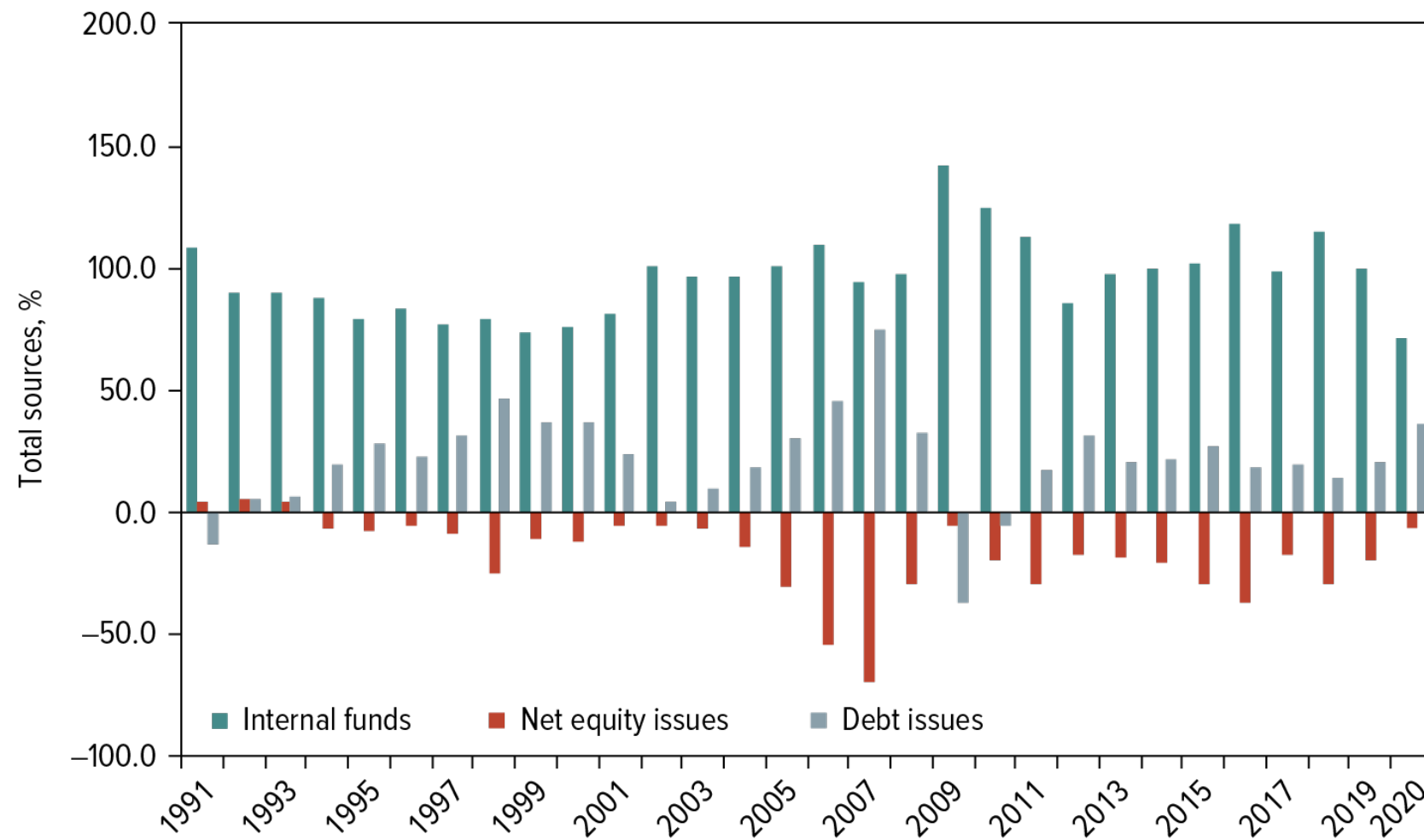
Patterns of corporate financing

- Firms may raise funds from external sources or plow back profits rather than distribute them to shareholders.
- Shareholders generally accept plow back decisions, provided that cash is invested in positive-NPV projects. Every positive-NPV project outlay increases shareholder value.
- If internal cash flows are not sufficient to pay for all investments, a company faces a financial deficit.
- To cover a deficit, a company must cut back on dividends in order to increase retained earnings, or raise new (debt or equity) capital.
- Financial management: What fraction of profits should be plowed back into the business rather than paid out to shareholders? What fraction of the financial deficit should be met with debt or equity?

Patterns of corporate financing (II)

- On average, internal funds (retained earnings plus depreciation) cover most of the cash needed for investments.
- Managers could be tempted to forgo a risky but positive NPV-project if it involves a new stock issue.
- Reason: Announcement of new equity issues are generally associated with negative stock returns (signaling).
- The mix of debt and equity financing varies widely from industry to industry and from firm to firm.
- Debt ratios also vary over time for particular firms. There is no constant / target debt ratio, and if there were, it would be time-varying.

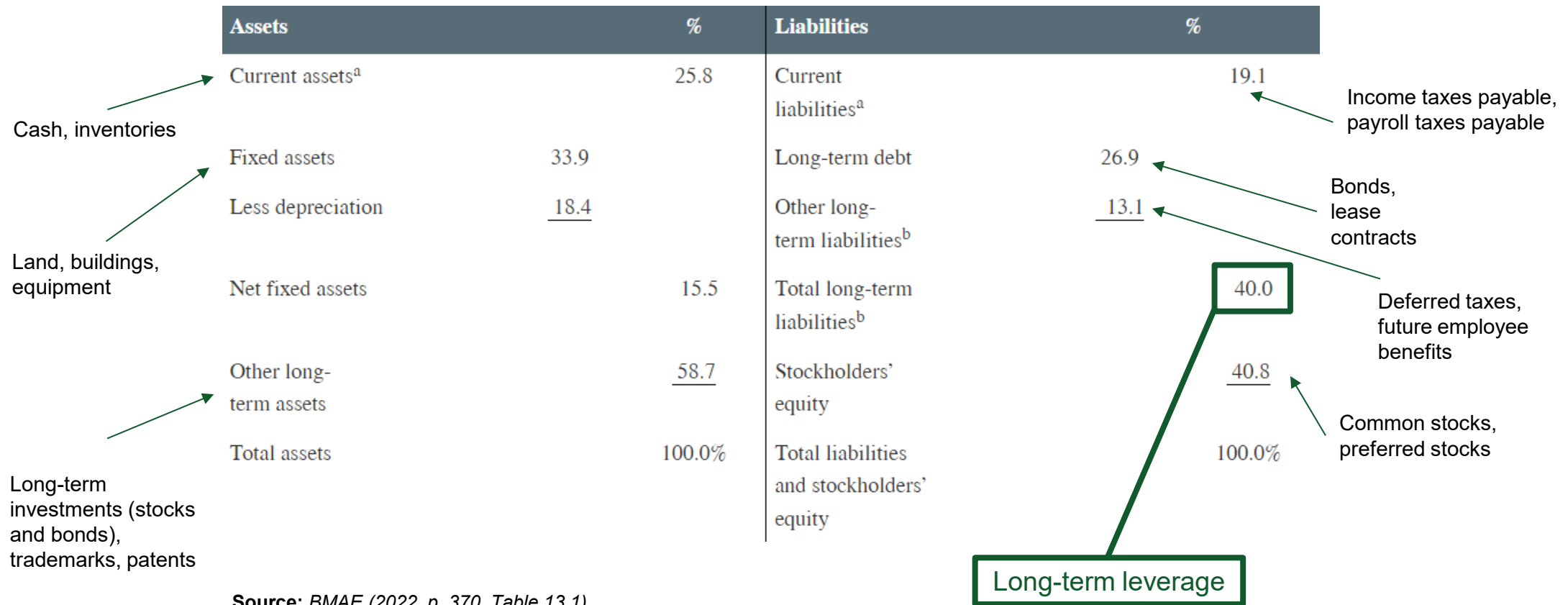
Sources of funds for U.S. non-financial corporations



Source: BMAE (2022, p.369, Figure 13.1)

Sources of funds for U.S. non-financial corporations (II)

- The aggregate balance sheet of all U.S. manufacturing corporations, if they were merged into a single firm (fourth quarter 2020; in percent of total assets):



Basics: Equity securities – Common stock

- Common stocks (often called equity securities) represent an **ownership** share in a publicly-held corporation.
- Each share of common stock entitles its owner to **cash flow rights** (according to the face / nominal value) plus (often one) **voting rights** on any matters of corporate governance, M&As, etc. at the corporation's annual shareholder meeting.
- A corporation sometimes issues **two classes of common stock**, one bearing the right to vote, the other not (generally trading at a lower price), or with differing voting rights.
- One important characteristic of common stock is the “**residual claim**” it offers, i.e., stockholders are last in line of all investors who have a claim on a firm's assets (and income).

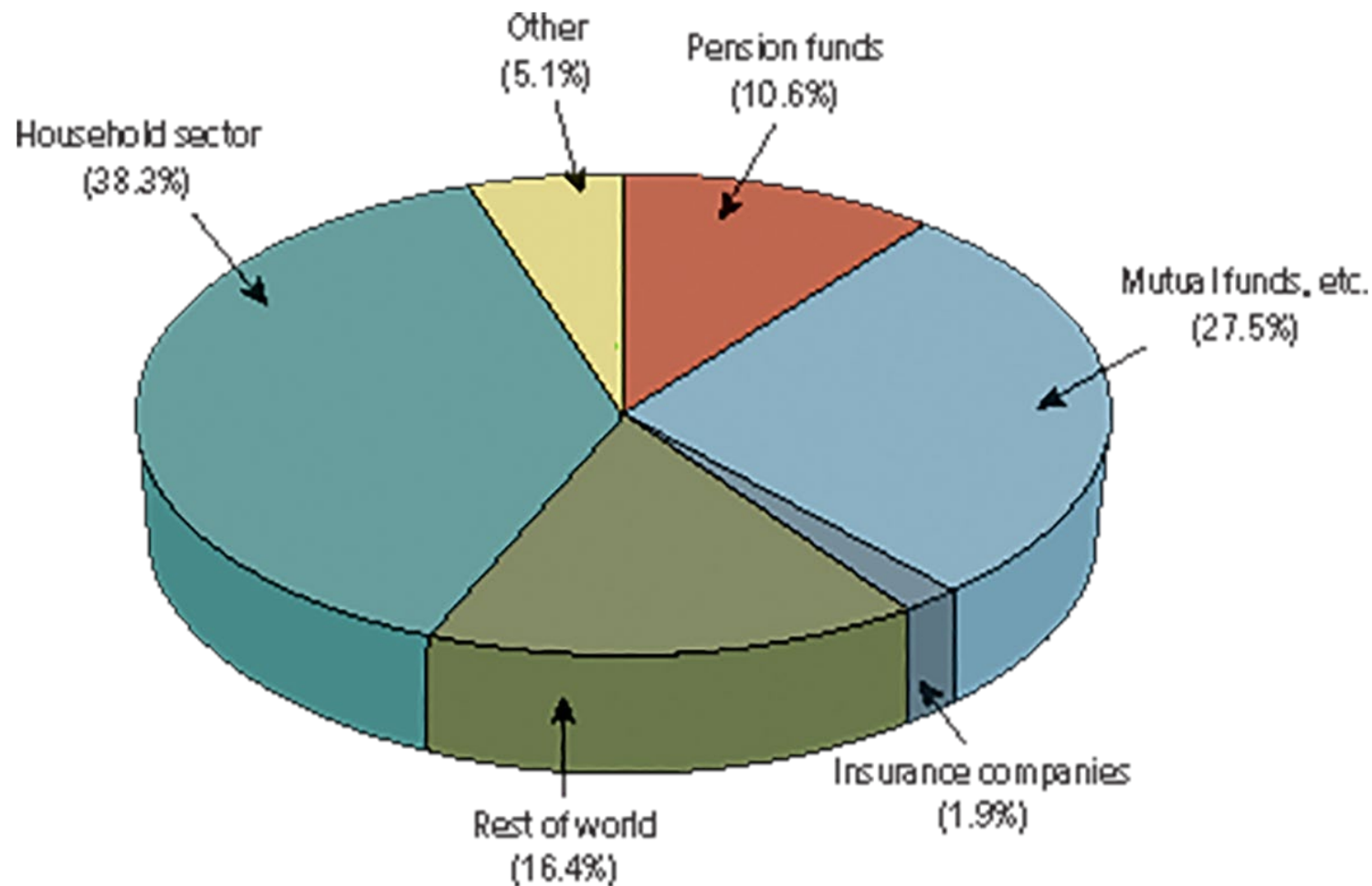
Basics: Equity securities – Common stock (II)

- Shareholders have **limited liability**: In the event of failure, they can lose their original investment; but they are not personally liable for the firm's obligations.
- Common stocks of most large corporations can be bought / sold on **stock exchanges**.
 - **Primary market**: new securities, e.g., initial public offerings (IPO) and seasoned equity offerings (SEO).
 - **Secondary market**: previously issued securities.
- Stocks of public firms which are not publicly traded are said to be “**closely held**”.

Basics: Equity securities – Preferred stock

- Preferred stocks have features similar to **both equity and debt**.
- Like a bond, it promises to pay its holder **a fixed amount of income** every year and it does not convey voting power.
- The firm has no contractual obligation to pay dividends to the preferred stockholders. However, **unpaid dividends are cumulated** and must be paid in full before any dividends may be paid to common stockholders.
- Preferred stocks **rank after bonds** in terms of the priority of their claims to the assets in the event of bankruptcy.
- As preferred stock payments are treated as dividends rather than interest, they are **not tax-deductible expenses** for the firm.

Holdings of corporate equities issued in the U.S. by U.S. and foreign corporations, December 2020



Source: BMAE (2022, p. 372, Figure 13.4)

*Rest of world refers to the holdings of U.S. issues by foreign residents

Voting procedures

- For many U.S. Companies, the entire board of directors is up for reelection each year.
- Some U.S. companies have **classified** (or **staggered**) boards, where only a third of the directors are reelected annually.
- On many issues a **simple majority** of shareholder votes is sufficient; however, the company charter may specify decisions that require a **supermajority** (e.g., 75%) of those eligible to vote, e.g., to approve a merger or change the charter.
- **Proxy contests** involve the firm's existing management and directors to compete with outsiders for effective control of the corporation:
 - A growing number of activist investors campaign for changes in management policy.
 - If they gain sufficient shareholder support, the firm may get the message without entering a proxy fight (e.g., Dan Loeb and Nestlé).

Event 1: Activist investor Dan Loeb's impact on Nestlé

- In June 2017, activist investor Dan Loeb acquired a \$3.5 billion stake in Nestlé (i.e., ca. 40 million shares or **1.25% of equity**) and became the company's 6th-largest shareholder; the company then moved to adopt most of his reforms.
- Dan Loeb criticized Nestlé's slow sales growth, declining stock price, and overly complex organizational structure, which resulted from a failure to divest businesses that did not fit into the “nutrition health and wellness” strategy.
- Dan Loeb put the blame on the corporate structure, and in particular the board of directors in charge of the company's strategic direction.
- Dan Loeb demanded in a letter in June 2017 reiterated in July 2018 that Nestlé:
 - sells its stake in several companies (e.g., Nestlé's stake in L'Oreal) and spins off businesses that do not fit into its strategy,
 - reorganizes into three divisions (beverages, nutrition, and grocery), and
 - adds an outside industry expert (food and beverage business) to the board.

Event 1: Does it pay to be an activist?

Dan S. Loeb, founder and CEO of Third Point; estimated wealth: \$3.5bn



Event 2: Joe Kaeser's effort to save Siemens

- Siemens CEO Joe Kaeser spend his time as a CEO (2013-2021) working to keep activist investors at bay and escaped becoming a target to activist investors.
- As activist shareholders often push companies to sell businesses, Joe Kaeser spent his years as CEO simplifying the structure of the conglomerate.
- Joe Kaeser implemented the following changes at Siemens:
 - Bolstering of its software business; acquisition of CD-adapco in 2016 for almost \$1 billion, and the \$4.5 billion buy of Mentor Graphics Corp.
 - Carve out of the health-care division, Siemens Healthineers, in March 2018; World's largest healthcare IPO on record with a deal value of \$4.5bn.
 - Adoption of a more holding-like structure that gives autonomy to units, while still being centrally managed.
 - Creation of an "ownership culture" under which employees buy shares and get a stake in the company's future.

The seniority of debt and debtholders' rights

- Debt has first claim on cash flows, but its claim is limited to interest payments and the repayment of a face value.
- The claim structure (or debt waterfall payment):
 - If default occurs, **senior** debt is first in line to be repaid.
 - **Junior**, or **subordinated**, debtholders are paid only after all senior debtholders are (fully!) satisfied.
 - However, all debtholders rank ahead of the preferred and common stockholders.
- In contrast to equity, debt does not have residual cash-flow rights and does not participate in the upside of a business.

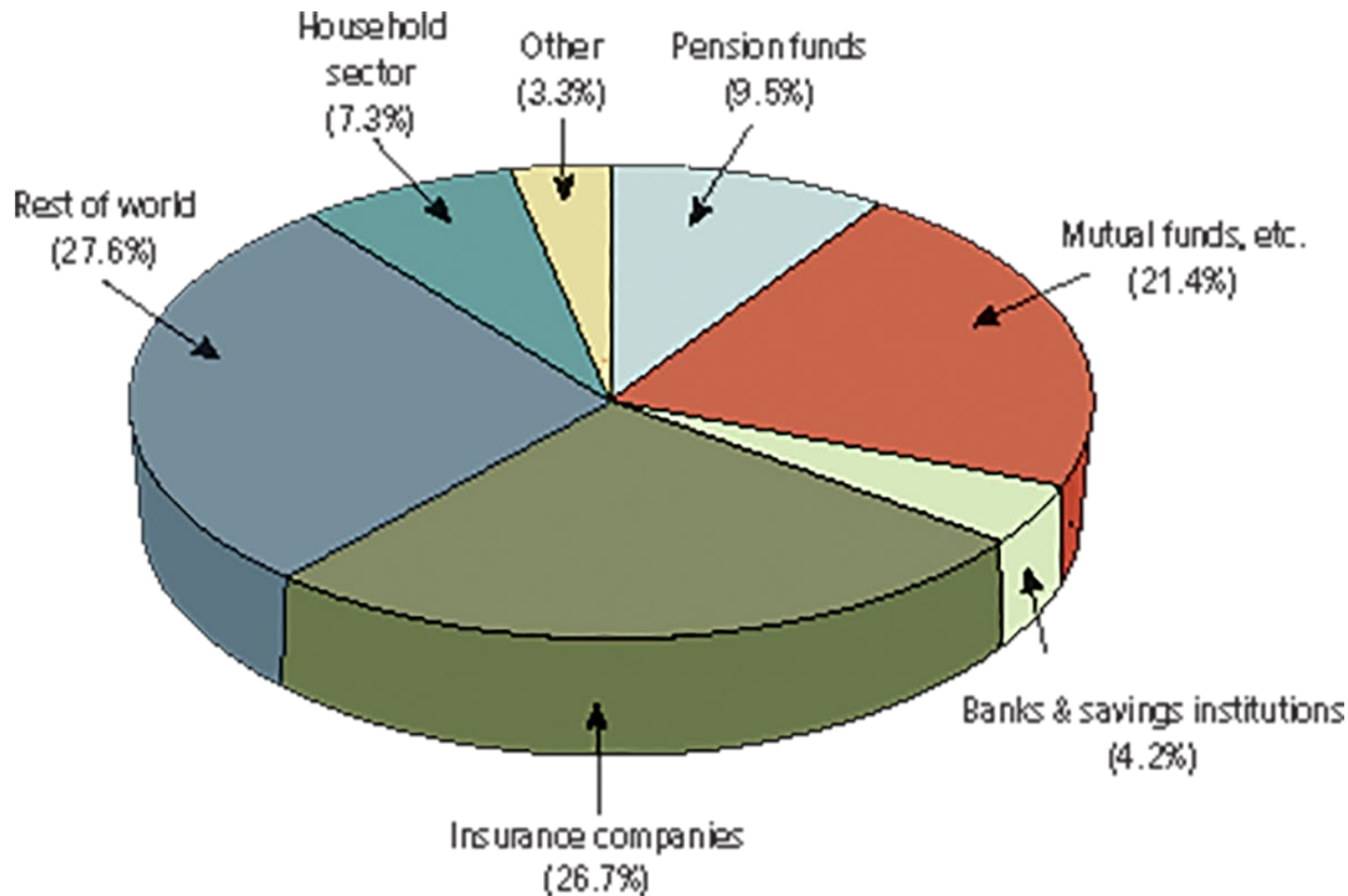
The seniority of debt and debtholders' rights (II)

- Because lenders are not considered to be owners of the firm, they do not normally have any voting power.
- Unlike equity, debt offers **no control rights** unless the firm defaults or violates debt covenants.
- However, owners are protected by **bankruptcy laws** and **credit covenants** (see corporate governance class).
- The firm also provides **insurance** to the lender that it will not take unreasonable risks, e.g., the borrower may agree to limit the extra amount of debt that it can issue, accept limitations on mergers, etc.
- In order to protect particular creditors, the firm may set aside some of its assets in terms of **collateral**: the debt is then **secured**.

Convertible debt

- Firms often issue securities that give the owner an option to **convert** them into other securities. These options may have a substantial effect on the value of the securities / bonds.
- A **warrant** gives the right to purchase a set number of the company's shares at a set price before a set date. Warrants and bonds are often sold together as a «package».
- A **convertible bond** gives the owner the option to exchange the bond for a predetermined number of equity shares.
 - The convertible bondholder hopes that the issuing company's share price will increase so that the bond can be converted at a profit.
 - If the share price decreases, there is no obligation to convert.

Holdings of bonds issued in the U.S. by U.S. and foreign corporations, December 2020



Source: BMAE (2022, p. 376, Figure 13.6)

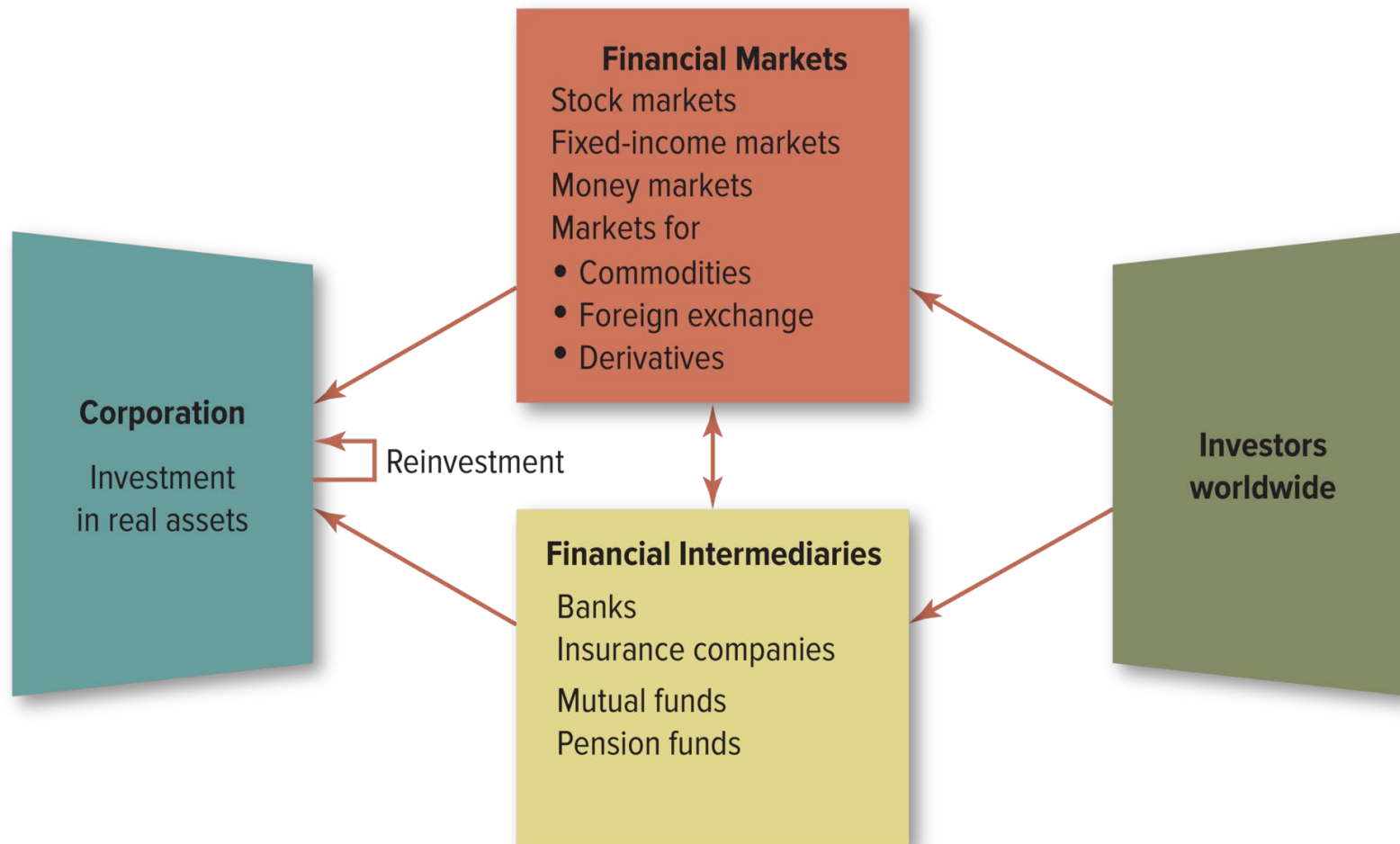
*Rest of world refers to the holdings of U.S. issues by foreign residents

Debt by another name

Some obligations are similar to debt, but treated differently in the accounts:

- Accounts payable:
 - Goods received, not yet paid for
 - They are essentially short-term debt
- Rents and leases:
 - Instead of borrowing money to buy new machinery, firms may lease the machinery, leading to similar (lease) payments as in case of a loan
- Unfunded obligations
 - Are similar to senior debt; an example are employee pension obligations
- Special-purpose entities (SPEs)
 - Raise cash through equity and debt issues of the SPE
 - They do not show up on balance sheet of the company

Flow of savings to investment



Source: BMAE (2022, p. 382, Figure 13.7)

Financial assets of intermediaries in the United States, December 2020

	\$ Billions
Mutual funds	\$19,563
Money market funds	4,336
Closed-end funds	279
ETFs	5,449
Hedge funds ^a	2,700
Pension funds	25,666
Banks and savings institutions	23,454
Insurance companies	12,279

Source: BMAE (2022, p. 383, Table 13.2)

Venture Capital

- Venture capital (VC) is a type of funding (equity investments) in young, growing private companies.
- VC investors are investment institutions or wealthy individuals, who finance firms before they can obtain debt or go public.
- VC firms pool funds from a variety of investors to build high risk financial portfolios (i.e., VC funds).
- Venture capitalists provide funding, and often also managerial services / advice, to the startup company in exchange for equity.
- Venture capitalists rarely give a young company up front all the money it will need; they define stages / milestones at which they provide funding to reach the next stage / milestone.
- For an example of a VC investment, see “Marvin example” in BMAE.

The role of venture capital funds

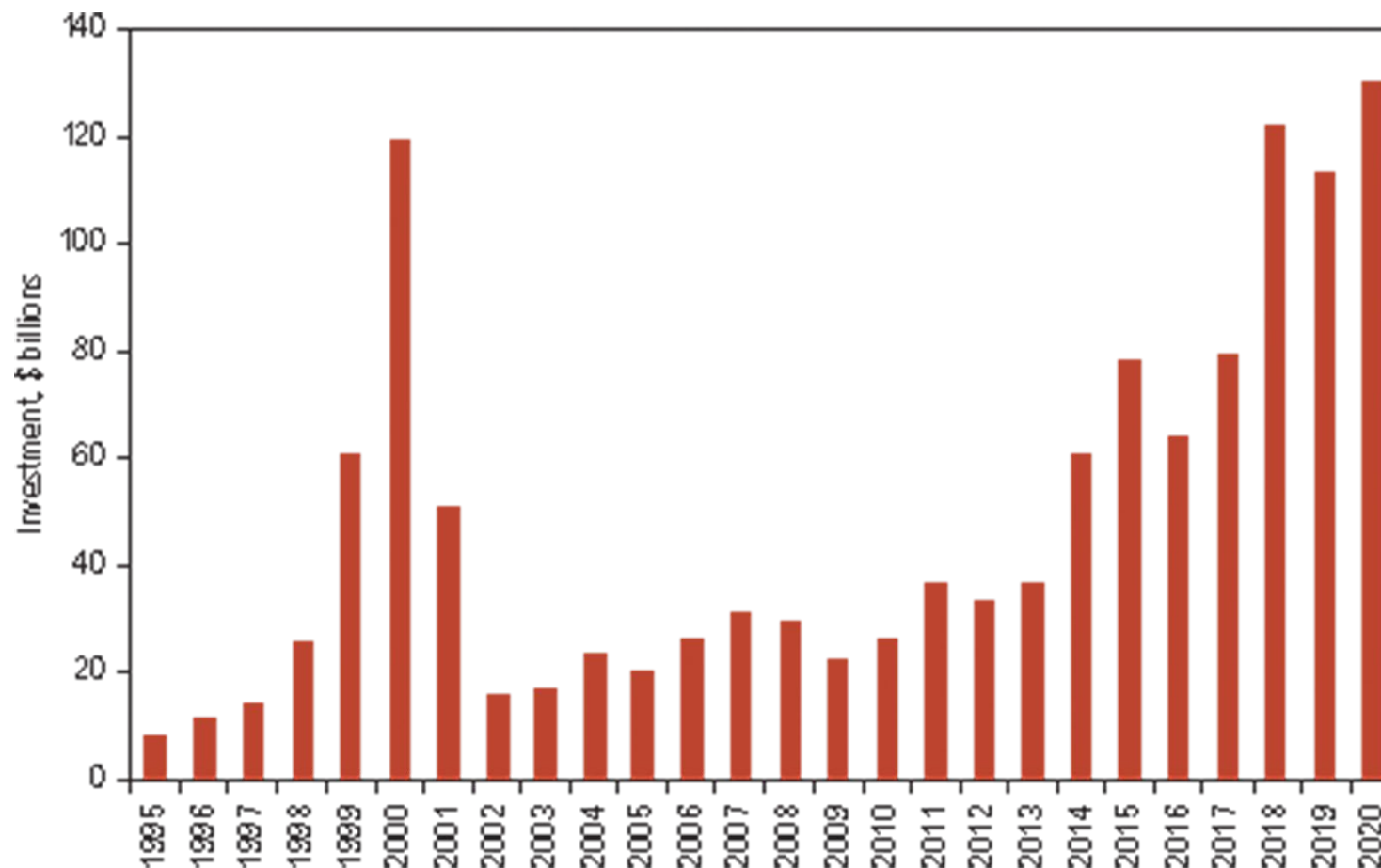
- Most VC funds are organized as limited private partnerships with a fixed life of about 10 years. The «limited partners» are the investors (e.g., pension funds).
- The management company (general partner) is responsible for making and overseeing the investments and, in return, receives a fixed fee and a share of the profits, called the «carried interest».
- A typical arrangement would be for the management company to receive a fee of 2% plus 20% of the profit (if any).

The role of venture capital funds (II)

- VC firms tend to specialize in young, often high-tech, firms that are difficult to evaluate but present the chance of becoming a big, public company in a profitable market.
- VC firms monitor these firms closely, provide ongoing advice and often play a major role in recruiting the senior management team.
- Venture capitalists may cash out on their investment in 2 ways:
 - Sell out to a larger firm.
 - IPO of the target company. This allows the VC fund to «cash out» and the original entrepreneurs to stay in control. About 50% of companies going to public have been backed by a VC company.

Venture capital investment in the United States

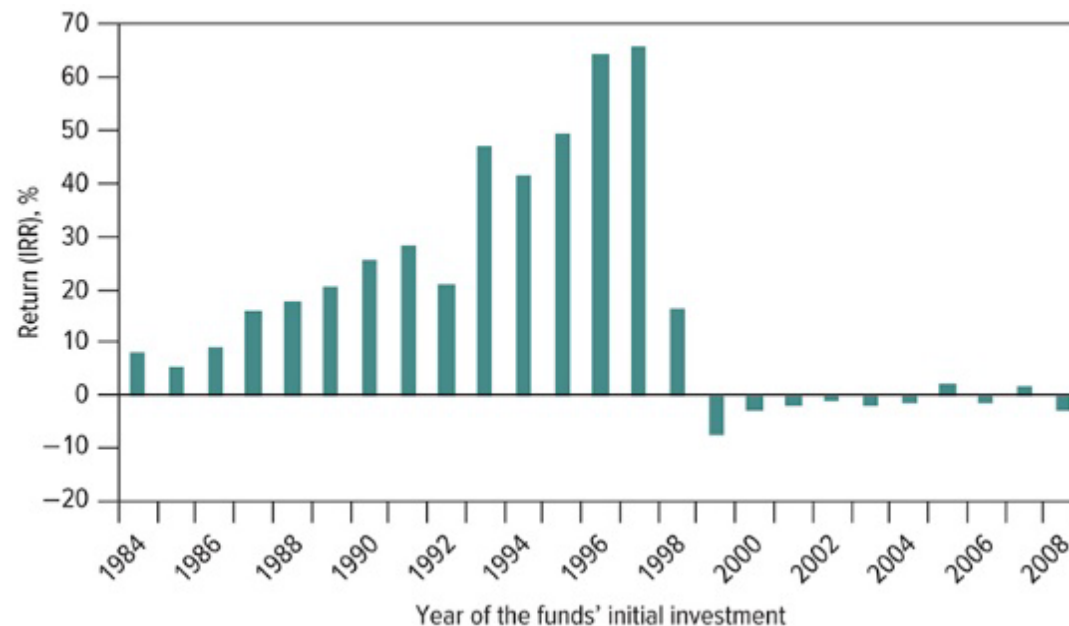
- VC funds invested nearly \$120 billion in 2000; two years later the figure had fallen to \$16 billion.



Source: BMAE (2022, p. 403, Figure 14.1)

Returns to venture capital investments

- The average annual internal rate of return (IRR) on venture capital funds was about 17% compared to an annualized S&P 500 return (dividends reinvested) of 9.6% over the period 1984-2008.
- Funds formed before 1998 earned dreamy returns, whereas most of the funds formed later made losses:



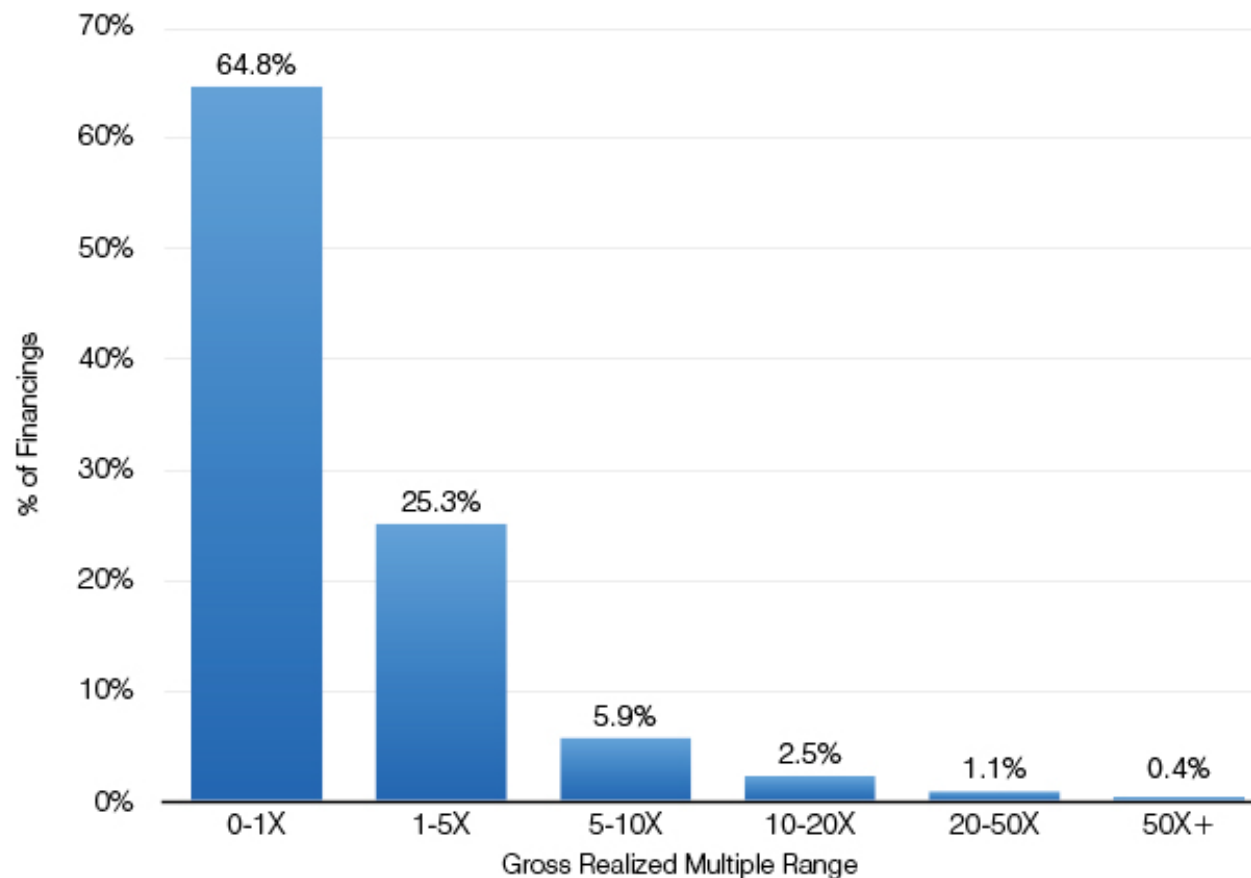
Source: BMA (2019, p. 395, Figure 15.2)

Performance of venture capital investments

- A problem in assessing the performance of VC funds is risk:
 - There is only one observation per fund: This does not allow the estimation of market-based risk (e.g., using a factor model).
 - The portfolio firms are not listed: Hence, the beta(s) of a VC fund cannot be derived from the weighted average beta of the portfolio firms' betas.

Returns to venture capital investments (II)

Distribution of U.S. Venture Returns
Based on 21,640 financings of companies that went out-of-business, were acquired, or went public in the 2004-2013 time period.

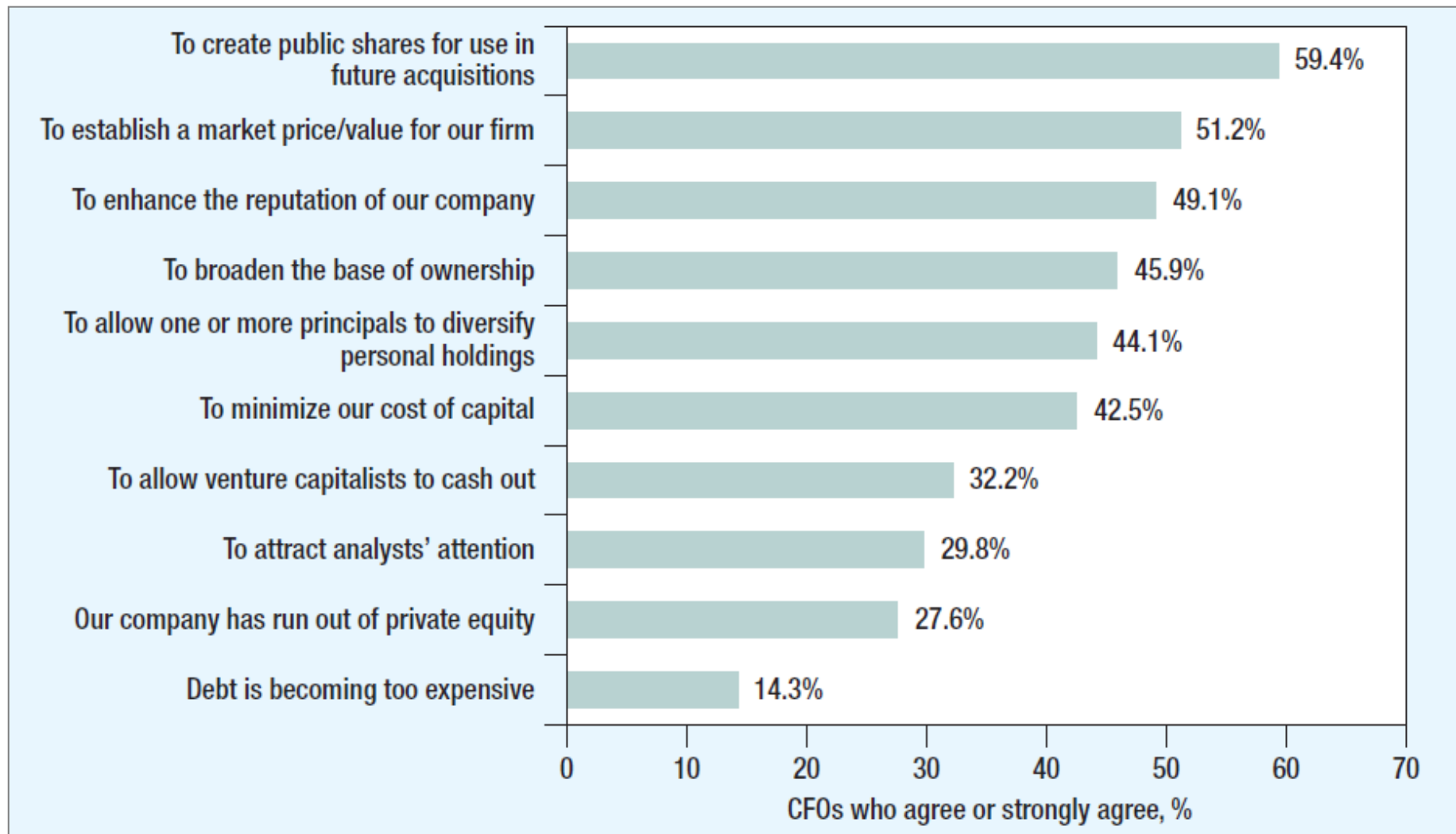


Source: Correlation Ventures based on data from Dow Jones VentureSource and other sources

Initial Public Offering (IPO) – Terminology

- **Initial Public Offering (IPO):** First offering of stock to the general public.
- **Underwriter:** Investment bank that buys an issue of securities from a company and resells it to the public.
- **Spread:** Difference between public offer price and price paid by underwriter.
- **Prospectus:** Formal summary that provides information on an issue of securities.
- **Underpricing:** Issuing securities at an offering price set below the market value of the security (i.e., the first-day stock price at the exchange).

Survey evidence on the motives for going public



Source: BMAE (2022, p. 405, Figure 14.2)



The underwriters

- **Underwriters:** The underwriters generally are the leading investments banks (e.g., Goldman Sachs, JP Morgan, BoA/ML, UBS, Deutsche Bank, etc.)
- **Firm Commitment:** underwriters buy the securities from the firm and then resell them to the public
- **Best-Efforts Commitment:** underwriters agree to sell as much of the issue as possible but do not guarantee the sale of the entire issue
- **Flotation Costs:** the costs incurred when a firm issues new securities to the public
 - What are some of the specific costs incurred when a firm issues new securities? (e.g., spread and underpricing)

Examples: Spreads and costs of new issues

Example 1:

Assume the issuing company incurs \$1 million in expenses to sell 3 million shares at \$40 each to an underwriter; the underwriter sells the shares at \$43 each. What is the spread for this deal?

Example 2:

How much will a firm receive in net funding from a “firm commitment” underwriting of 250,000 shares priced to the public at \$40 if a 10% underwriting spread has been added to the price paid by the underwriter to the issuer? Additionally, the firm pays \$600,000 in legal fees.

Underpricing of IPOs

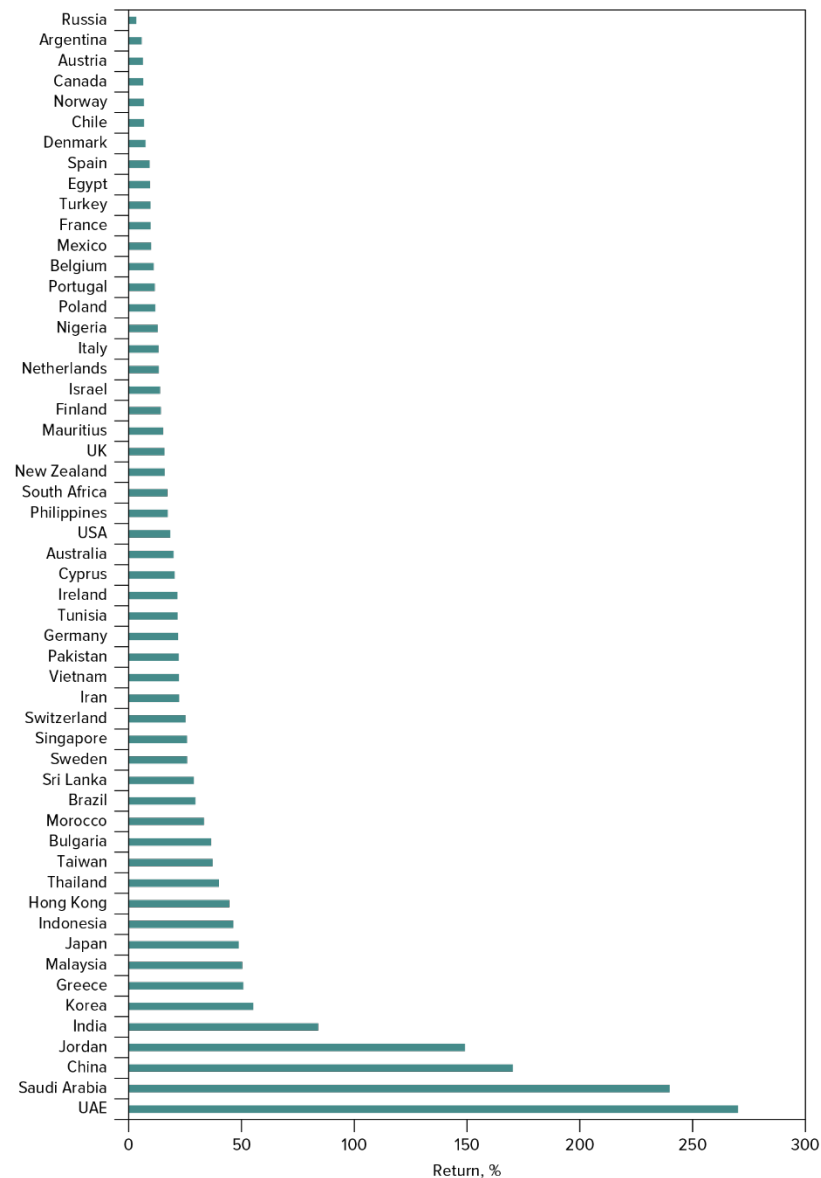
Example 3:

Assume the issuer incurs \$1 million in other expenses to sell 3 million shares at \$40 each to an underwriter and the underwriter sells the shares at \$43 each. By the end of the first day's trading, the issuing company's stock price had risen to \$70. What is the total underpricing in USD?

Two real examples of underpricing:

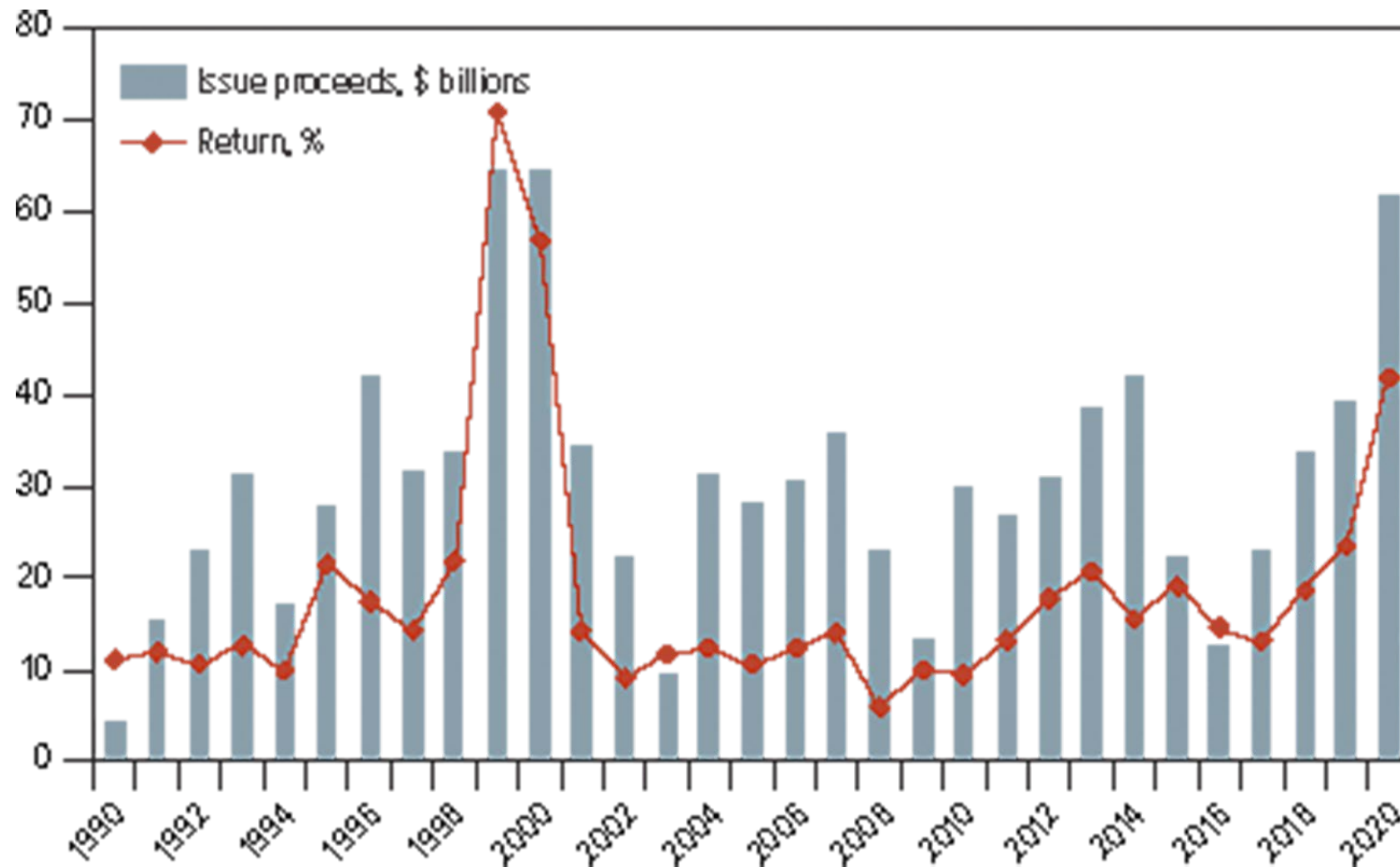
- 1) **Akamai Technologies**, founded in 1998, went public on October 29, 1999, at an offer price of USD 26 and with a first day closing price of USD 145.19 (+558%) leaving USD 1,072,687,500 on the table.
- 2) **Goldman Sachs** went public on May 4, 1999, at an offer price of USD 53 and with a first day closing price of USD 70.38 (+33%) leaving USD 959,100,000 on the table.

Average initial returns from investing in IPOs in different countries



Source: BMAE (2022, p. 412, Figure 14.3)

IPO proceeds in the U.S. and average first-day returns, 1990-2020



Source: BMAE (2022, p. 413, Figure 14.4)

Why is there underpricing?

- Does underpricing compensate for risk? This would mean that riskier IPOs are more underpriced.
- Are (some) IPOs underpriced on purpose to avoid the negative reputation of a “bad start” in the public market?
- Incentives and analyst coverage: Underwriter with influential “star” analyst issuing positive reports may compensate for underpricing.
- Spinning, i.e., the use of underpriced IPOs as a currency to compensate for other business. Underwriters started to set up personal brokerage accounts for venture capitalists and the executives of issuing firms in order to allocate hot IPOs to them (“spinning”) in exchange for other business.

The long-term performance of IPO stocks

IPO long-term returns (3Y) by company size (sales over last 12 months):

Sales (in 2021\$)	Number of IPOs	Average First-day Return	Average 3-year Buy-and-hold Return		
			IPOs	Market-adjusted	Style-adjusted
0-9.999 mm	1,714	21.8%	-7.5%	-45.5%	-30.3%
10-19.999 mm	692	27.9%	-0.9%	-39.3%	-18.0%
20-49.999 mm	1,488	24.1%	19.5%	-23.3%	-7.9%
50-99.999 mm	1,392	18.0%	30.7%	-13.1%	-3.0%
100-499.999 mm	2,290	14.0%	45.9%	3.2%	9.0%
500 mm and up	1,199	9.8%	37.5%	0.3%	0.0%
0-99.999 mm	5,286	22.2%	11.0%	-29.9%	-15.2%
100 mm and up	3,489	12.6%	43.0%	2.2%	5.9%
1980-2020	8,775	18.4%	23.7%	-17.1%	-6.8%

Source: Ritter (2022)

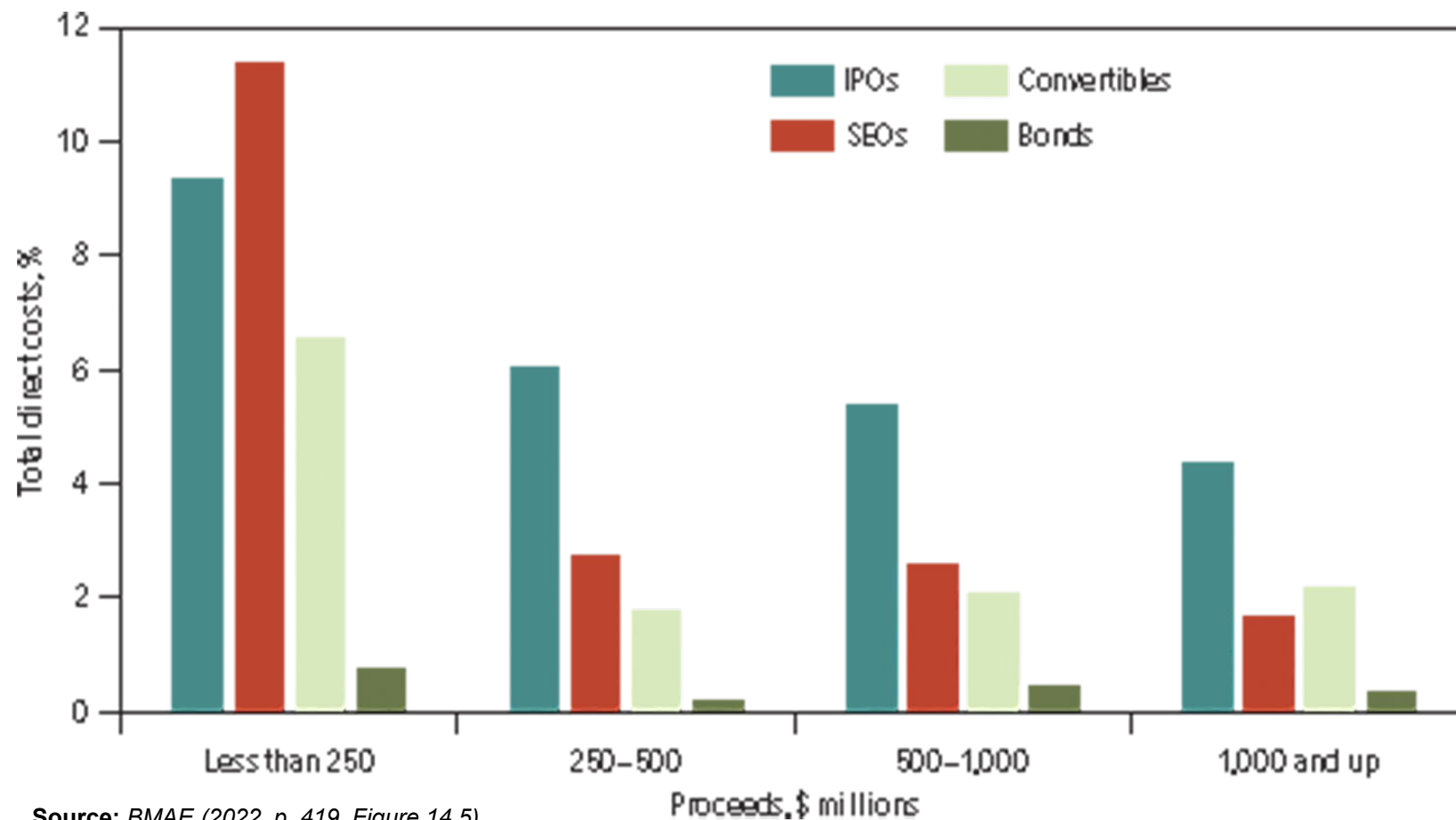
The long-term performance of IPO stocks (II)

Some observations on IPO long-term performance:

- The 8,775 U.S. IPOs taking place between 1980 and 2020 have a positive average return of 23.7% over the first three years.
- However, as compared to the stock market (CRSP value-weighted index) the IPO firms underperform by 17.1%.
- As compared to a portfolio of size- and book-to-market matched firms, IPO firms underperform by 6.8%.
- Smaller IPOs perform worse than larger IPOs.
- VC-backed IPOs perform better than non-VC-backed IPOs (not in the table).
- There are IPO waves: The largest to date was in the 1990s. Since then IPO levels are much lower than they were before.

Total direct costs as a percentage of gross proceeds

- Study of total issue costs (underwriter spreads plus administrative costs) for several thousand issues between 2014 and 2019.



Source: BMAE (2022, p. 419, Figure 14.5)

Key takeaways

1. Most of the money for new investments comes from profits that companies retain and reinvest. The remainder comes from selling new debt or equity securities.
2. Lenders and stockholders have different cash–flow rights and also different control rights.
3. VC investments usually take place through VC funds.
4. VC funds performed very well before the burst of the dotcom bubble, but underperformed the stock market on average after.
5. IPOs are heavily underpriced on average; there are several potential explanations for this observed pattern.
6. In the long-term, IPOs generally do not outperform common benchmarks, if anything, they underperform.

Literature

Mandatory readings

- Brealey, R., Myers, S., Allen, F., and A. Edmans, 2022, Principles of Corporate Finance, 14th Edition, New York: McGraw-Hill, Chapter 13 (An Overview of Corporate Financing) & Chapter 14 (How Corporations Issue Securities).

Additional readings

- Harris, R.S., Jenkinson, T., and S.N. Kaplan, 2014, Private equity performance: What do we know, Journal of Finance 69, 1851-1882.
- Brau, J.C., and S.E. Fawcett, 2006, Evidence on what CFOs think about the IPO process: Practice, theory and managerial implications, Journal of Applied Corporate Finance 18, 107-117.